

PVCM — Prediction Version Continuity Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

PRCS — Prediction Record & Continuity Standard

Operational Layer: Prediction Record & Historical Continuity

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Version Continuity

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

PVCM governs how material changes to a Governed Prediction are represented as historically connected prediction versions while preserving Prediction Record identity, temporal order, prior-state visibility, change context, and lineage.

It governs the transition:

Governed Initial Prediction Record → Material Prediction Change → New Historical Prediction Version → Preserved Version Continuity

Its foundational principle is:

A new prediction version may change the prediction. It must not erase the version that existed before it.



Canonical Definition

The Prediction Version Continuity Module (PVCM) defines the operational governance requirements by which material changes to a Governed Prediction become distinct, identifiable, temporally ordered, historically connected prediction versions without breaking Prediction Record identity or silently replacing prior states.

PVCM governs:

Version Eligibility; Material Change Determination; Version Creation; Version Identity; Parent-Version Reference; Version Lineage; Version Timestamp; Effective Time; Change Classification; Change Description; Change Source; Change Reason; Prediction Content Change; Probability Change; Uncertainty Change; Horizon Change; Boundary Change; Assumption Change; Prediction Basis Change; Methodology Change; Predictor / Model Change; Condition Change; Scenario Change; version branching; version merging; correction versions; rollback; reinstatement; version conflict; continuity qualification; version traceability.

Its primary output is:

Governed Prediction Version Continuity Record

Operational Role

PVCM is the second operational module within PRCS.

Its focused operational question is:

When a prediction changes materially, how should the new state be represented so that the change is historically explicit, the prior state remains reconstructable, and both states remain connected to the same governed Prediction Record where appropriate?

PVCM does not decide whether a prediction should be revised.

PMRS owns revision governance.

PVCM governs what happens to historical version continuity once a material change exists.

Prediction Revision Decision $\neq$ Prediction Version Continuity

Internal Module Flow

Prediction Record Formation → Prediction Version Continuity → Prediction History Preservation → Prediction Record State Transition → Prediction Record Re-Entry

Current Position:

Prediction Version Continuity

1. Prediction Version

A Prediction Version is an identifiable historical state of a Governed Prediction within a continuing Prediction Record.

Examples:

P1

P2

P3

Each version represents what the prediction meant during a defined historical state.

2. Version Continuity

Version Continuity is the governed relationship linking one prediction state to the next.

Conceptually:

P1 → P2 → P3

The sequence should remain reconstructable.

3. Version Eligibility

A new version should be created where a change materially affects predictive meaning or historical interpretation.

Possible states:

New Version Required

New Version Recommended

Technical Change Only

No Material Version Change

Unable to Determine

4. Material Change

A Material Change is a change capable of affecting:

what is predicted; probability; uncertainty; Prediction Horizon; Predictive Boundary; assumptions; Prediction Basis; methodology; conditions; interpretation; reliance; outcome comparison; performance assessment.

5. Technical Change

A purely technical change may not require a new substantive prediction version.

Examples may include:

formatting; metadata correction with no predictive meaning change; storage migration; non-semantic encoding change.

Technical Change ≠ Predictive Version Change by Default

6. Materiality Determination

Materiality should be determined based on governance effect rather than cosmetic difference.

The question is:

Would this change alter how the prediction should be interpreted, relied upon, compared, or evaluated historically?

If yes:

→ a new version is generally required.

7. Version Identity

Each material prediction version should be uniquely identifiable within the Prediction Record.

8. Stable Record Identity

Version creation should not automatically create a new Prediction Record.

New Version ≠ New Prediction Record by Default

9. Version Number

An implementation may use:

1.0

1.1

2.0

or another versioning convention.

PVCM does not mandate one numbering format.

10. Version Sequence

The sequence should preserve historical order.

Example:

P1 → P2 → P3

A later version should not be presented as having existed earlier.

11. Parent-Version Reference

Each new version should identify the prior version from which it directly evolved where applicable.

12. Version Lineage

Version lineage preserves:

which version came from which prior version.

This becomes essential where history branches.

13. Linear Versioning

A simple prediction lifecycle may remain linear:

P1 → P2 → P3 → P4

14. Branching Versioning

A prediction may branch into scenario-specific versions:

P2 → P3A

P2 → P3B

The branch should remain explicit.

15. Branch Identity

Each branch should preserve:

common origin; branching point; branch-specific predictive state.

16. Branching ≠ Duplicate Prediction by Default

Branches may represent genuinely distinct predictive states rather than redundant copies.

17. Version Merge

Separate prediction branches may later be consolidated.

Where this occurs, the merged state should preserve references to its contributing version lineage.

18. Merge ≠ Historical Collapse

A merge should not erase the branch history that preceded it.

19. Version Timestamp

Each version should preserve when it became historically identifiable.

20. Effective Time

The version creation timestamp may differ from the time at which the new version becomes operationally effective.

Where material:

Version Creation Time $\neq$ Version Effective Time

21. Publication Time

A version may be created before publication.

Publication time should remain separate where relevant.

22. Communication Time

A version may be communicated later than it was created.

This distinction can matter for reliance history.

23. Revision Time

Where PMRS triggered revision, the applicable revision event time should remain linkable.

24. Change Classification

Version change may be classified as:

Content Change

Probability Change

Uncertainty Change

Horizon Change

Boundary Change

Assumption Change

Basis Change

Methodology Change

Condition Change

Scenario Change

Predictor / Model Change

Composite Change

25. Content Change

A prediction's core statement may change.

Example:

"Event X will occur."

becomes:

"Event X will probably not occur."

This is clearly material.

26. Probability Change

Example:

35% → 62%

The prior 35% state must remain historically preserved.

27. Uncertainty Change

Example:

High uncertainty → Moderate uncertainty

This may materially affect interpretation.

28. Confidence Change

Where confidence is separately governed:

Confidence Change ≠ Probability Change

29. Horizon Change

Example:

Q4 2028 → Q2 2029

This should normally create a new material prediction version.

30. Horizon Extension

Extending a horizon after the event fails to occur within the original horizon should not rewrite the original version.

Horizon Extension ≠ Historical Horizon Replacement

31. Horizon Contraction

Shortening the prediction horizon may also materially change predictive meaning.

32. Boundary Change

Example:

EU market

becomes:

Germany only

The scope change must remain version-specific.

33. Geographic Change

Geographic changes should preserve the original geography.

34. Population Change

Changes in predicted population should remain explicit.

35. Product or Object Change

A version should not silently change the underlying Prediction Object.

A sufficiently different object may require a new Prediction Record rather than a new version.

36. Version vs New Prediction Boundary

A key PVCMM determination is whether change creates:

Same Prediction, New Version

or:

New Prediction Object

This distinction should remain governed.

37. Assumption Change

A material change in assumptions may justify a new prediction version.

38. Basis Change

New information or evidence may change the Prediction Basis.

The resulting prediction may become a new version.

39. Basis Addition

Adding new material information should remain traceable.

40. Basis Removal

Removing material information should also remain explicit.

41. Methodology Change

A new Prediction Methodology may materially change the basis of the prediction.

42. Methodology Version Change

Example:

Methodology v2 → v3

The prediction version should preserve which methodology generated each state.

43. Methodology Change ≠ Historical Rewrite

A new methodology cannot be retrospectively applied to earlier prediction states unless a separate retrospective analysis is clearly distinguished.

44. Model Change

A new AI or forecasting model may generate a new version of the prediction.

45. Model Change and Prediction Identity

A model change does not automatically mean a new Prediction Record.

The governing question is whether the underlying Prediction Object remains continuous.

46. Predictor Change

A human or organizational predictor may change.

Where the prediction remains part of the same governed predictive record, the new predictor context should remain explicit.

47. Condition Change

Conditional predictions may require new versions when material conditions change.

48. Scenario Change

Scenario changes may create:

new version; new branch; new Prediction Record.

The appropriate structure depends on predictive continuity.

49. Change Source

PVCM should preserve what triggered the change where available.

Examples:

new information; assumption change; model update; methodology update; human judgment; operational event; external condition change; regulatory event.

50. Change Reason

Where governed, the record should preserve why the prediction changed.

51. Known Reason vs Unknown Reason

A version should distinguish:

Reason Known

from:

Reason Not Recorded

and:

Unable to Determine

52. Change Description

A concise representation of what changed should remain available.

Example:

Probability increased from 42% to 71% after supplier shutdown confirmation.

53. Change Diff

Machine-readable systems may preserve a structured difference between versions.

54. Semantic Diff

A semantic difference may be more important than literal text difference.

Example:

Small wording change can create major scope change.

55. Text Change ≠ Meaning Change

Minor text differences may be non-material.

56. Meaning Change Without Large Text Change

A single changed word may materially alter predictive meaning.

PVCM should govern semantic materiality.

57. Version Freeze

Once a new version becomes historical, it should not be silently rewritten.

58. Historical Version Immutability

A historical version may receive corrections or annotations.

The original state should remain reconstructable.

59. Version Correction

Where a version contains a recording error, a correction may be applied through an explicit correction relationship.

60. Correction Version

A Material Correction may require a specific correction version.

61. Correction ≠ Predictive Revision

A correction fixes the historical record.

A predictive revision changes the prediction itself.

These should remain distinct.

62. Rollback

A system may return operationally to an earlier prediction state.

Example:

P3 → rollback to P2

This does not erase P3.

63. Rollback State

A rollback should preserve:

rolled-back version; restored version; rollback time; reason.

64. Rollback ≠ Historical Deletion

P3 remains part of the record even if P2 becomes active again.

65. Reinstatement

A superseded or inactive version may later be reinstated.

The reinstatement should create a new state transition rather than pretending the intervening history never occurred.

66. Reinstatement ≠ Original Version Never Changed

67. Version Conflict

Two systems may claim different current versions.

PVCM should preserve the conflict rather than silently selecting one.

68. Concurrent Versioning

Different authorized environments may temporarily hold distinct prediction versions.

Concurrency should remain explicit where material.

69. Canonical Version

An implementation may designate one version as operationally canonical at a given time.

This does not erase non-canonical historical versions.

70. Canonical Version ≠ Only Historical Version

71. Duplicate Version

Multiple identical copies of one prediction version should not automatically be treated as distinct predictive versions.

72. Duplicate Copy ≠ New Version

73. Version Provenance

Each material prediction version should preserve sufficient provenance.

Potential elements include:

creating actor or system; predecessor version; change source; methodology; model; timestamp; source record.

74. Version Integrity

Version history should be protected against:

silent insertion; silent deletion; backdating; version reordering; predecessor substitution; metadata falsification.

75. Version Sequence Integrity

A system should be able to determine the legitimate sequence of versions.

76. Backdated Version

If a version is entered after the fact, the distinction between:

effective historical claim time

and:

record insertion time

should remain visible.

77. Retrospective Version

A version reconstructed later from historical evidence should be explicitly marked as retrospective where appropriate.

78. Retrospective Version ≠ Contemporaneous Version

79. Version Completeness

Each version should preserve enough context to remain meaningful independently within its lineage.

80. Minimal Version Sufficiency

A material version should normally preserve:

version ID; predecessor; content state; effective time; change classification; applicable predictive context.

81. Version Qualification

A version may carry qualifications such as:

Incomplete

Retrospective

Timestamp Approximate

Source Disputed

Methodology Reference Missing

82. Version Status

Possible version states may include:

Active

Historical

Superseded

Withdrawn

Rolled Back

Reinstated

Disputed

83. Active Version

At a given operational time, one or more versions may be active depending on governance structure.

84. Historical Version

A version becomes historical when it is no longer the only current state or when its time-bounded status has passed.

85. Superseded Version

Supersession changes current operational status.

It does not change historical existence.

86. Withdrawn Version

A withdrawn version remains part of the Prediction Record.

87. Version Comparison

PVCM may support structured comparison among versions to show how the prediction evolved.

88. Version Drift

Version Drift describes the cumulative change of the prediction across multiple revisions.

Example:

P1: 20%

P5: 90%

The historical trajectory may matter later.

89. Revision Frequency

Frequent version changes may be historically significant.

PVCM preserves them.

Interpretation belongs elsewhere.

90. Version Compression Risk

Systems may be tempted to store only:

First Version

and:

Final Version

This can erase material intermediate history.

First + Final $\neq$ Complete Version History by Default

91. Governance Requirements

A governed PVCM implementation should be able to:

1. determine Version Eligibility; 2. determine Material Change; 3. distinguish Technical from Predictive Change; 4. establish new Prediction Versions; 5. preserve stable Prediction Record identity; 6. assign Version Identity; 7. preserve Version Sequence; 8. preserve Parent-Version Reference; 9. preserve Version Lineage; 10. support Linear Versioning; 11. support Branching Versioning; 12. preserve Branch Identity; 13. support Version Merge; 14. preserve Merge History; 15. preserve Version Timestamp; 16. distinguish Version Creation Time from Effective Time; 17. preserve Publication Time where material; 18. preserve Communication Time where material; 19. preserve Revision Time; 20. classify Change Type; 21. preserve Content Change; 22. preserve Probability Change; 23. preserve Uncertainty Change; 24. distinguish Confidence Change from Probability Change; 25. preserve Horizon Change; 26. prevent Horizon Extension from rewriting historical horizon; 27. preserve Boundary Change; 28. preserve geographic and population changes; 29. distinguish Version from New Prediction Object; 30. preserve Assumption Change; 31. preserve Prediction Basis Change; 32. preserve Basis Addition and Removal; 33. preserve Methodology Change; 34. preserve Methodology Version; 35. prevent methodology updates from rewriting earlier versions; 36. preserve Model Change; 37. preserve Predictor Change; 38. preserve Condition Change; 39. preserve Scenario Change; 40. preserve Change Source; 41. preserve Change Reason; 42. distinguish Known from Unknown Change Reason; 43. preserve Change Description;
44. support structured Change Diff; 45. support Semantic Diff; 46. distinguish Text Change from Meaning Change; 47. preserve Version Freeze; 48. prevent silent historical rewriting; 49. support Version Correction; 50. distinguish Correction from Predictive Revision; 51. support Rollback; 52. preserve Rollback State; 53. prevent Rollback from deleting historical versions; 54. support Reinstatement; 55. preserve Reinstatement history; 56. identify Version Conflict; 57. support Concurrent Versioning; 58. support Canonical Version designation; 59. preserve non-canonical historical versions; 60. identify Duplicate Versions; 61. distinguish duplicate copies from new versions; 62. preserve Version Provenance; 63. preserve Version Integrity; 64. preserve Version Sequence Integrity; 65. identify Backdated Versions; 66. identify Retrospective Versions; 67. distinguish retrospective from contemporaneous versions; 68. assess Version Completeness; 69. preserve Minimal Version Sufficiency; 70. preserve Version Qualification; 71. preserve Version Status; 72. identify Active Version; 73. identify Historical Version; 74. identify Superseded Version; 75. identify

Withdrawn Version; 76. support Version Comparison; 77. preserve Version Drift; 78. preserve Revision Frequency; 79. prevent Version Compression from hiding material history; 80. preserve complete version traceability; 81. produce a Governed Prediction Version Continuity Record.

92. Governance Logic

PVCM applies the following operational logic:

Has the Governed Prediction materially changed?

If no:

→ no new predictive version is required.

If yes:

→ determine whether the change remains part of the same Prediction Record.

Does the underlying Prediction Object remain sufficiently continuous?

If yes:

→ create a new Prediction Version.

If no:

→ a new Prediction Record may be required.

What changed?

→ content;

→ probability;

→ uncertainty;

→ horizon;

→ boundary;

→ assumptions;

→ Prediction Basis;

→ methodology;

→ model;

→ predictor;

→ conditions;

→ scenario.

Which prior version does the new version derive from?

→ preserve Parent-Version Reference.

When was the version created and when did it become effective?

→ preserve both where materially different.

What triggered the change?

→ preserve source and reason where available.

Is the change a correction or a predictive revision?

→ distinguish explicitly.

Did branching occur?

If yes:

→ preserve branch structure.

Did rollback, reinstatement, or merge occur?

If yes:

→ preserve the state transition without deleting intermediate versions.

Is the version retrospective?

If yes:

→ preserve retrospective qualification.

Then:

→ establish Version Identity;

→ preserve Lineage;

→ preserve Diff;

→ preserve Version Status;

→ preserve Provenance;

→ preserve Integrity;

→ pass complete version continuity into Prediction History Preservation.

Canonical transition:

Material Prediction Change → New Historical Version → Linked Version Lineage → Preserved Version Continuity

93. Minimum Implementation Framework

A. Prediction Record ID

Persistent reference to the continuing Prediction Record.

B. Version ID

Unique identifier for the new material prediction version.

C. Parent-Version Reference

Immediate predecessor or predecessor set.

D. Version Timestamp

Time the version became historically identifiable.

E. Effective Time

Time the version became operationally applicable where different.

F. Prediction State Snapshot

Complete material predictive state for the version.

G. Change Classification

Probability, uncertainty, horizon, boundary, assumption, basis, methodology, model, condition, scenario, content, or composite change.

H. Change Description

Human- or machine-readable explanation of the material change.

I. Change Source

Information, actor, system, event, methodology, or process triggering the change.

J. Methodology / Model Context

Applicable predictive methodology and model versions.

K. Version Lineage

Predecessor, branch, merge, or derived relationships.

L. Version Status

Active, Historical, Superseded, Withdrawn, Rolled Back, Reinstated, Disputed, or equivalent.

M. Version Qualification

Retrospective, incomplete, disputed, approximate, corrected, or other material qualification.

N. Version Provenance

Creator, system, source, timestamps, transformation, and record origin references.

O. Version Integrity Record

Integrity state, correction history, conflict state, and traceability information.

94. Governance Outputs

PVCM may produce:

Governed Prediction Version Continuity Record Version ID Parent-Version Reference Version Lineage
Version Timestamp Effective Time Prediction State Snapshot Change Classification Change Description
Change Source Change Reason Probability Change Record Uncertainty Change Record Horizon Change
Record Boundary Change Record Assumption Change Record Prediction Basis Change Record
Methodology Change Record Model Change Record Predictor Change Record Condition Change Record
Scenario Change Record Branch Record Merge Record Rollback Record Reinstatement Record Version
Conflict Record Version Qualification Version Status Version Provenance Version Integrity State Version
Traceability Record

The primary output is:

Governed Prediction Version Continuity Record

This becomes governed input to:

PHPM — Prediction History Preservation Module

95. Critical Distinctions

Prediction Revision Decision $\neq$ Prediction Version Continuity

Technical Change $\neq$ Predictive Version Change by Default

New Version $\neq$ New Prediction Record by Default

Version Creation Time $\neq$ Version Effective Time

Confidence Change $\neq$ Probability Change

Horizon Extension $\neq$ Historical Horizon Replacement

Methodology Change $\neq$ Historical Rewrite

Text Change $\neq$ Meaning Change

Correction $\neq$ Predictive Revision

Rollback ≠ Historical Deletion

Reinstatement ≠ Original Version Never Changed

Canonical Version ≠ Only Historical Version

Duplicate Copy ≠ New Version

Retrospective Version ≠ Contemporaneous Version

First + Final ≠ Complete Version History by Default

Prediction Lineage ≠ Prediction Identity

Version Continuity ≠ Version Immutability Without Correction

96. Enterprise Application

PVCM can govern prediction version continuity across:

AI forecasting systems; enterprise forecasting; financial forecasting; investment research; prediction markets; industrial prediction; predictive maintenance; supply-chain forecasting; public-sector forecasting; technology forecasting; risk forecasting; expert forecasting; Human-AI systems; autonomous agents; multi-agent systems; continuously updating predictive models.

Its function remains constant:

allow predictions to change without allowing the historical record of those changes to disappear.

97. Use Case 1 — A 25% Forecast Quietly Becomes 80% and the Platform Shows Only the Latest Number

A forecasting platform predicts:

Probability of Event X: 25%

Two months later:

41%

Later:

63%

Three days before resolution:

80%

Event X occurs.

The platform subsequently displays only:

80% — Correct

The critical governance question is:

Was the predictor an 80% predictor throughout the prediction lifecycle, or did the prediction evolve from 25% to 80% as new information arrived?

PVCM preserves:

P1 — 25%

P2 — 41%

P3 — 63%

P4 — 80%

with:

timestamps; predecessor references; change reasons; relevant Prediction Basis changes; applicable methodology or model versions.

The final 80% state remains legitimate.

But it cannot become the historical substitute for P1–P3.

Final Version $\neq$ Complete Version History

New Version $\neq$ Historical Replacement

98. Use Case 2 — An AI System Changes Forecast Scope Instead of Admitting the Original

Prediction Missed Its Horizon

An AI system predicts:

“Technology X will achieve full commercial deployment across the European Union by 31 December 2028.”

December 2028 arrives without full EU deployment.

The system then changes the prediction to:

“Technology X will achieve commercial deployment in at least one EU market by 31 December 2029.”

If the original record is overwritten, later observers may see only the second statement and assume the prediction was always broader in time and narrower in geography.

PVCM identifies material changes:

Prediction Horizon: 31 December 2028 → 31 December 2029

Predictive Boundary: Full EU deployment → At least one EU market

These changes require a new historical prediction version.

The lineage becomes:

P1 — Full EU deployment by 31 Dec 2028

→

P2 — At least one EU market by 31 Dec 2029

P1 remains available for POFPS performance assessment.

P2 becomes the new predictive state.

The critical governance question is:

Can extending time and narrowing scope convert an earlier failed predictive claim into the same prediction?

PVCM answers:

The prediction may remain within the same historical lineage if continuity is legitimate, but the change must be explicit and versioned.

Horizon Extension $\neq$ Historical Horizon Replacement

Boundary Change $\neq$ Original Prediction Rewrite

99. Canonical Boundary

PVCM governs the creation, identity, sequencing, lineage, temporal anchoring, classification, provenance, integrity, branching, merging, correction, rollback, reinstatement, conflict handling, and traceability of material Prediction Versions within a continuing Prediction Record. It ensures that changes to content, probability, uncertainty, horizon, Predictive Boundary, assumptions, Prediction Basis, methodology, model, predictor, conditions, scenarios, or other material predictive states create historically explicit version relationships rather than silently replacing prior states. It does not decide whether a prediction should be revised, form the original Prediction Record, preserve the complete lifecycle history beyond version continuity, govern broader record- state transitions, or determine lifecycle re-entry.

PRFM owns:

Prediction Record Formation

PVCM owns:

Prediction Version Continuity

PHPM subsequently owns:

Prediction History Preservation

PRSTM subsequently owns:

Prediction Record State Transition

PRRM subsequently owns:

Prediction Record Re-Entry

100. Canonical Closing Statement

A prediction that changes is not a governance problem. A prediction that changes without leaving a trace is.

PVCM preserves every material predictive state as part of a connected historical lineage so that probability shifts, scope changes, new assumptions, model upgrades, and horizon extensions cannot silently replace what was previously predicted.

A new version may become the present prediction. It must never become a substitute for the past.

[↑](#)

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